

ASTU MSJ SUMMER BOOTCAMP 2025

BOOTCAMP AND FULL-STACK PROJECT DOCUMENTATION

1. INTRODUCTION

The **ASTU MSJ Summer Bootcamp 2025** is a structured educational and skills-development program designed to equip Muslim students with strong foundations in **Competitive Programming (CP)** and **Full-Stack Development**.

The bootcamp started in **Summer 2025** and provides students with a combination of technical education, practical experience, mentorship, teamwork, programming contests, mini-projects, experience-sharing sessions, and progress tracking.

The program is designed to go beyond traditional classroom learning. Students are expected to learn concepts, practice them regularly, work with their teammates, communicate with mentors, participate in contests, complete weekly tasks, and apply their knowledge through practical projects.

The bootcamp is founded on the belief that knowledge and technical skills are not merely academic achievements. The skills developed through the program are intended to become a means of benefiting Islam, the Muslim community, and humanity at large.

As part of the Full-Stack Development phase, students develop a real-world application called the **ASTU MSJ Bootcamp Management System**. The system is designed to digitally manage important bootcamp activities, including student applications, admission, mentorship, attendance, assignments, grading, announcements, and progress tracking.

ASTU MSJ BOOTCAMP MANAGEMENT SYSTEM

The project is designed to solve an actual need within the bootcamp by providing a centralized platform for managing students, mentors, applicants, attendance, assignments, announcements, and progress.

The project combines the knowledge gained during the Full-Stack Development phase and applies it to a real-world application.

2. ASTU MSJ BOOTCAMP MANAGEMENT SYSTEM

The **ASTU MSJ Bootcamp Management System** is a web-based platform designed to manage the bootcamp's activities from the application stage through the student's learning journey.

The system connects three main categories of users:

- Students
- Mentors
- Administrators

The system provides different functionality to each role according to their responsibilities.

The main areas of the system include:

- Landing page
- Applicant registration
- Application review
- Applicant filtering
- Interview management
- Student admission
- Authentication
- Student management
- Mentor management
- Group management
- Attendance
- Progress tracking
- Assignments
- Assignment submission
- Assignment grading
- Announcements
- Administrative overview

3. PURPOSE OF THE BOOTCAMP MANAGEMENT SYSTEM

The purpose of the system is to centralize and digitize important bootcamp processes.

The platform helps reduce dependence on separate documents, spreadsheets, and informal communication channels.

The system provides a single environment where:

- Applicants can apply.
- Administrators can review applicants.
- Successful applicants can become students.
- Students can log in.
- Students can view their progress.
- Students can view their attendance.
- Students can receive assignments.
- Students can submit assignments.
- Students can receive grades.
- Students can receive announcements.
- Mentors can monitor their assigned students.
- Mentors can record attendance.
- Mentors can grade assignments.
- Administrators can monitor the overall bootcamp.

4. SYSTEM USERS

The system has three primary roles.

4.1 Student

Students are the main learners participating in the bootcamp.

4.2 Mentor

Mentors are responsible for guiding and monitoring assigned students.

4.3 Administrator

Administrators are responsible for managing the overall system and bootcamp operations.

The basic structure is:

ADMIN



Manages Bootcamp

MENTOR



Guides Assigned Students

STUDENT



Participates in Bootcamp

5. LANDING PAGE

The landing page is the public entry point of the Bootcamp Management System.

It introduces visitors to the ASTU MSJ Summer Bootcamp and provides information about the program.

The landing page can communicate:

- Bootcamp objectives
- Learning tracks
- Competitive Programming
- Full-Stack Development
- Mentorship
- Contests
- Mini-projects
- Experience-sharing sessions
- Final project
- Bootcamp journey

The landing page provides navigation to important actions such as:

- Applying to the bootcamp
- Logging into the system
- Learning more about the program

The purpose of the landing page is to give visitors a clear understanding of the bootcamp before they decide to apply.

6. APPLICANT REGISTRATION

The system provides a registration page for individuals who want to apply to the bootcamp.

A new applicant can complete an application form and submit their information.

The application process is separate from the student login system.

Submitting an application does not immediately create an active student account.

Instead, the applicant enters the selection process.

The process is:



7. APPLICATION REVIEW AND FILTERING

After an applicant submits their application, the application is reviewed.

The administrators evaluate the submitted information according to the bootcamp's selection requirements.

Applicants who meet the initial requirements are filtered and moved to the interview phase.

Applicants who do not meet the required criteria may not proceed to the interview.

This provides a structured selection process.

The purpose of filtering is to ensure that the interview stage focuses on applicants who successfully pass the initial application review.

8. INTERVIEW PROCESS

Applicants who pass the initial filtering stage proceed to the interview phase.

The interview provides an opportunity for the bootcamp team to further evaluate applicants.

After the interview, the applicant receives an outcome.

The applicant can either:

- Pass the interview
- Not pass the interview

Applicants who pass the interview can be accepted as students.

9. STUDENT ACCEPTANCE

When an applicant successfully passes the selection and interview process, they are accepted into the bootcamp as a student.

At this stage, the applicant becomes an active student within the Bootcamp Management System.

A student account is created for the accepted applicant.

The system then provides the student with credentials that allow them to access the platform.

10. TEMPORARY PASSWORD AND EMAIL

When a student is accepted, the system generates a temporary password.

The temporary password is sent to the student's registered email address.

The student can then log in using:

- Their registered email
- The temporary password

The overall process is:

Accepted Applicant



Student Account Created



Temporary Password Generated



Temporary Password Sent by Email



Student Logs In



Student Dashboard

This provides a controlled transition from applicant to active student.

11. LOGIN AND AUTHENTICATION

The system provides a centralized login page.

Students, mentors, and administrators can log into the system using their credentials.

After successful authentication, the system identifies the user's role and provides access to the appropriate dashboard.

The login process can be represented as:

User



Enter Email and Password



Authentication



Role Identification



Appropriate Dashboard

Students are directed to the Student Dashboard.

Mentors are directed to the Mentor Dashboard.

Administrators are directed to the Admin Dashboard.

12. ROLE-BASED ACCESS

Different users have different responsibilities.

Therefore, users should not have access to every function of the system.

For example:

- A student can view their own progress.
- A mentor can view assigned students' progress.
- An administrator can view overall progress.
- A mentor can record attendance for assigned students.
- An administrator can monitor attendance across the bootcamp.
- A student can submit assignments.
- A mentor can grade assignments.
- An administrator can manage assignments.

This role-based structure helps maintain security and keeps the interface relevant to each user.

13. STUDENT DASHBOARD

The Student Dashboard is the main workspace for students.

After logging in, students can access important information about their bootcamp participation.

The dashboard provides access to:

- Student profile
- Progress
- Attendance
- Announcements
- Assignments
- Assignment status
- Other relevant bootcamp information

The dashboard is designed to give students a central location where they can monitor their learning journey.

14. STUDENT PROGRESS

Students can view their progress through the system.

The bootcamp tracks progress in two major areas:

1. Competitive Programming
2. Development

The progress information helps students understand their current learning status.

Students can use the progress section to determine:

- What they have completed.
- What they still need to complete.
- Their progress in CP.
- Their progress in Development.
- Areas that may require more effort.

Progress tracking encourages students to take responsibility for their own learning.

15. STUDENT ATTENDANCE

Students can view their attendance records.

Attendance is an important part of bootcamp participation.

The system allows students to monitor their attendance and understand their participation over time.

Attendance information can help students recognize when they are missing sessions and encourage them to maintain consistent participation.

16. STUDENT ANNOUNCEMENTS

Students can access announcements published through the system.

Announcements provide a centralized communication mechanism.

Announcements can contain:

- Important reminders
- Schedule information
- Deadline information
- Contest information
- Project updates
- General bootcamp notices
- Other important information

Students can therefore receive important bootcamp information directly through the platform.

17. STUDENT ASSIGNMENTS

Students can view assignments provided by mentors or administrators.

Assignment information may include:

- Assignment title
- Description
- Instructions
- Deadline
- Attached files or resources
- Submission requirements

The assignment section gives students a clear view of the tasks they are expected to complete.

18. ASSIGNMENT SUBMISSION

Students can submit completed assignments through the system.

The system can maintain the status of each assignment.

Possible assignment states include:

- Submitted
- Pending
- Overdue
- Upcoming Deadline

This allows students to understand which assignments require action and which tasks have already been completed.

19. MENTORSHIP SYSTEM

Mentorship is a central part of the ASTU MSJ Summer Bootcamp.

Each student group has **two mentors**:

1. Competitive Programming Mentor
2. Development Mentor

The two mentors have different technical responsibilities.

The CP mentor focuses on Competitive Programming.

The Development mentor focuses on Full-Stack Development.

This provides students with specialized guidance in both major areas of the bootcamp.

20. STUDENT GROUP STRUCTURE

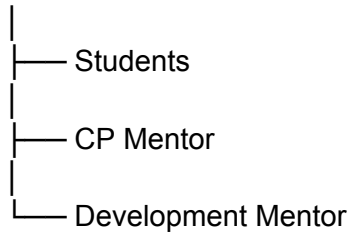
Students are organized into groups.

Each group has:

- Multiple students
- One CP mentor
- One Development mentor

The structure is:

Student Group



The group structure allows mentors to monitor and support the students assigned to them.

21. GENDER-BASED MENTORSHIP

The bootcamp follows a gender-based mentorship structure.

Female students are assigned female mentors, while male students are assigned male mentors.

This principle is considered when assigning mentors to students.

The mentorship structure therefore considers both:

- Technical specialization
- Gender-based mentor assignment

Within the appropriate mentor assignment, students receive support from:

- A CP mentor
 - A Development mentor
-

22. COMPETITIVE PROGRAMMING MENTOR

The CP mentor is responsible for supporting students in the Competitive Programming track.

The CP mentor can help students with:

- Problem solving
- Data Structures
- Algorithms
- Programming practice
- Contest preparation
- Difficult programming problems
- CP-related questions
- CP progress

The CP mentor can also monitor students' progress and identify areas where additional support may be needed.

23. DEVELOPMENT MENTOR

The Development mentor is responsible for supporting students in the Full-Stack Development track.

The Development mentor can help students with:

- HTML
- CSS
- JavaScript
- React
- Node.js
- Express.js
- MongoDB
- Authentication
- Next.js
- Development tasks
- Mini-projects
- Full-stack projects

The mentor can also monitor development progress and provide technical guidance when students face challenges.

24. EXPERIENCE-SHARING SESSIONS

Experience-sharing sessions are an important part of the bootcamp.

These sessions allow students to learn from people who have already experienced academic, technical, professional, or career-related journeys.

The sessions can cover topics such as:

- Software development
- Competitive programming
- Technology careers
- Academic experiences
- Career preparation
- Personal development
- Learning strategies
- Challenges and lessons learned

The bootcamp plans for **8 experience-sharing sessions**.

25. WEEKLY CONTESTS

Programming contests are an important part of the Competitive Programming phase.

The bootcamp plans for **9 contests**.

Contests allow students to apply the concepts they learn during CP sessions.

Through contests, students develop:

- Problem-solving speed
- Algorithmic thinking
- Time management
- Debugging skills
- Ability to work under pressure
- Competitive programming experience

Contest participation can also contribute to the student's CP progress.

26. WEEKLY MINI-PROJECTS

The bootcamp plans for **7 mini-projects**.

Mini-projects allow students to apply development concepts in practical situations.

Instead of learning development only through lectures, students build small applications or features.

Mini-projects help students develop:

- Coding skills
- Project-building skills
- Debugging ability
- Practical development experience
- Time management
- Teamwork

The mini-projects also prepare students for the larger final project.

27. SOFT SKILLS AND TEAMWORK

The bootcamp recognizes that technical knowledge alone is not sufficient for successful software development.

Students therefore develop soft skills alongside their technical skills.

These include:

- Communication
- Teamwork
- Presentation skills
- Collaboration
- Time management
- Responsibility
- Leadership
- Problem solving
- Professional communication

Teamwork is especially important during the final full-stack project.

28. STUDENT PROGRESS TRACKING

The bootcamp provides a progress sheet that allows students to monitor their development.

Progress is mainly tracked in:

- Competitive Programming
- Development

The progress sheet allows students to see their learning journey and understand which activities they have completed.

It also allows mentors to monitor their students and provide additional guidance when necessary.

29. CP PROGRESS

The CP progress section can include information related to:

- Problems solved
- Data Structures
- Algorithms
- Contest participation
- CP tasks
- Other CP milestones

The purpose is to help students and mentors understand the student's progress in Competitive Programming.

30. DEVELOPMENT PROGRESS

The Development progress section can include:

- Completed development topics
- Weekly development tasks
- Mini-projects
- Frontend activities
- Backend activities

- Authentication
- Full-stack project milestones
- Advanced development topics

This allows students to track their journey from learning individual technologies to developing complete applications.

31. STUDENT EXPECTATIONS

Students are expected to actively participate in the bootcamp.

The major expectations are:

1. Attend all lectures.
2. Participate in all contests.
3. Complete weekly tasks on time.
4. Communicate with mentors.
5. Communicate with teammates.
6. Ask questions when they face difficulties.
7. Participate in team activities.
8. Track their own progress.
9. Contribute to projects.
10. Continuously work to improve their technical skills.

Students are expected to take responsibility for their own learning.

32. ADMIN DASHBOARD

The Admin Dashboard is the central management area of the system.

Administrators have the broadest level of access because they are responsible for managing the overall bootcamp.

The Admin Dashboard can provide access to:

- Applicants
- Students
- Mentors
- Student groups

- Mentor assignments
 - Attendance overview
 - Progress overview
 - Announcements
 - Assignments
 - User management
-

33. ADMIN APPLICANT MANAGEMENT

Administrators can manage applicants through the system.

They can:

- View submitted applications.
- Review applicant information.
- Filter applicants.
- Move selected applicants toward interviews.
- Update applicant status.
- Accept successful applicants.
- Manage unsuccessful applicants.

This provides a centralized digital admission workflow.

34. ADMIN STUDENT MANAGEMENT

Administrators can manage accepted students.

Student management can include:

- Viewing students
- Viewing student information
- Managing student accounts
- Organizing students into groups
- Assigning mentors
- Managing student status

This allows administrators to maintain an organized student database.

35. ADMIN MENTOR MANAGEMENT

Administrators can manage mentors within the system.

They can:

- Create mentor accounts.
- View mentor information.
- Manage mentor information.
- Assign mentors to groups.
- Connect mentors with students.
- Maintain the mentorship structure.

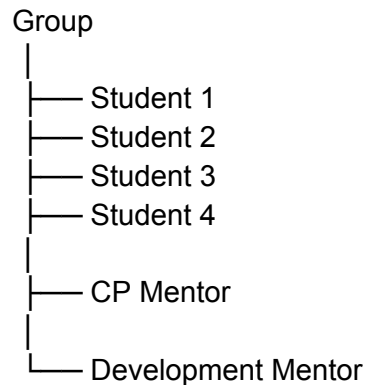
Mentor management is important because each group requires both a CP mentor and a Development mentor.

36. GROUP MANAGEMENT

The system allows administrators to organize students into groups.

A group contains students and their assigned mentors.

The structure can be represented as:



Group organization makes it easier to manage:

- Mentorship
- Attendance
- Progress
- Assignments

- Communication
-

37. ATTENDANCE MANAGEMENT

Attendance is managed through the system.

Mentors can record attendance for their assigned students.

Students can view their own attendance records.

Administrators can view an overall attendance overview.

This creates three levels of attendance functionality:

Student:

View own attendance.

Mentor:

Record and monitor attendance of assigned students.

Admin:

View overall attendance information.

38. ADMIN ATTENDANCE OVERVIEW

The administrator can view attendance information across the bootcamp.

This provides a broader picture of student participation.

The attendance overview can help administrators identify:

- Students with low attendance
 - Groups with attendance issues
 - Overall participation
 - Attendance trends
 - Students who may require attention
-

39. MENTOR PROGRESS MONITORING

Mentors can view the progress of students assigned to them.

This allows mentors to understand:

- Student CP progress
- Student development progress
- Completed tasks
- Incomplete activities
- Areas requiring improvement

Progress monitoring allows mentors to provide more targeted guidance.

40. ADMIN PROGRESS OVERVIEW

Administrators can view progress information at a broader level.

The admin progress overview allows the bootcamp administration to monitor student development across the program.

This can include:

- Competitive Programming progress
- Development progress
- Assignment progress
- Task completion
- Other tracked activities

The administrator can therefore understand the overall performance of the bootcamp.

41. ANNOUNCEMENT MANAGEMENT

Administrators can create and publish announcements.

Announcements provide an official communication channel between the bootcamp administration and users.

Announcements may include:

- Schedule changes
- Important deadlines
- Contest reminders
- Project updates
- General notices
- Bootcamp activities
- Important instructions

Depending on the system's configuration, announcements can be targeted to different audiences.

42. ASSIGNMENT MANAGEMENT

Administrators can create and manage assignments.

An assignment may contain:

- Title
- Description
- Instructions
- Deadline
- Attached resources
- Submission requirements

Assignments can be made available to the appropriate students.

This allows assignments to be managed directly through the platform.

43. MENTOR ASSIGNMENT GRADING

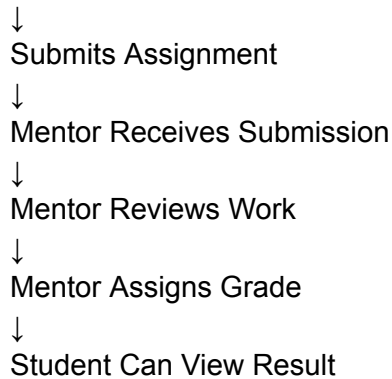
Mentors can review and grade assignments submitted by their assigned students.

The process is:

Student



Completes Assignment



Mentor grading provides students with feedback about their performance.

44. COMPLETE APPLICANT WORKFLOW

The complete application process is:

1. Visitor opens the landing page.
 2. Visitor learns about the bootcamp.
 3. Visitor opens the application/registration page.
 4. Applicant fills in the application form.
 5. Applicant submits the application.
 6. Administrator reviews the application.
 7. Administrator filters applicants.
 8. Selected applicants proceed to the interview stage.
 9. Applicant participates in an interview.
 10. Interview result is recorded.
 11. Successful applicants are accepted.
 12. A student account is created.
 13. A temporary password is generated.
 14. The temporary password is sent to the student's email.
 15. Student logs into the system.
 16. Student accesses the Student Dashboard.
-

45. COMPLETE STUDENT WORKFLOW

After acceptance, the student workflow is:

1. Receive temporary password through email.
 2. Log into the system.
 3. Access the Student Dashboard.
 4. View personal information.
 5. View assigned mentors.
 6. View CP progress.
 7. View Development progress.
 8. View attendance.
 9. View announcements.
 10. View assignments.
 11. Complete assignments.
 12. Submit assignments.
 13. View assignment status.
 14. View grades and feedback.
 15. Continue participating in the bootcamp.
 16. Track progress throughout the program.
-

46. COMPLETE MENTOR WORKFLOW

The mentor workflow is:

1. Mentor logs into the system.
 2. Mentor accesses the Mentor Dashboard.
 3. Mentor views assigned students.
 4. Mentor views student progress.
 5. Mentor records attendance.
 6. Mentor creates or manages assignments where permitted.
 7. Students complete assignments.
 8. Students submit assignments.
 9. Mentor reviews submissions.
 10. Mentor grades assignments.
 11. Mentor provides feedback where applicable.
 12. Mentor continues monitoring student progress.
-

47. COMPLETE ADMIN WORKFLOW

The administrator workflow is:

1. Administrator logs into the system.
 2. Administrator accesses the Admin Dashboard.
 3. Administrator reviews applicants.
 4. Administrator filters applicants.
 5. Administrator manages interview-stage applicants.
 6. Administrator accepts successful applicants.
 7. Student accounts are created.
 8. Temporary credentials are sent to accepted students.
 9. Administrator manages students.
 10. Administrator manages mentors.
 11. Administrator organizes student groups.
 12. Administrator assigns mentors.
 13. Administrator monitors attendance.
 14. Administrator monitors student progress.
 15. Administrator creates announcements.
 16. Administrator creates assignments.
 17. Administrator oversees the overall bootcamp.
-

48. RELATIONSHIP BETWEEN BOOTCAMP AND MANAGEMENT SYSTEM

The Bootcamp Management System directly represents the real structure and activities of the ASTU MSJ Summer Bootcamp.

Bootcamp Activity	System Feature
Student Application	Applicant Registration
Application Review	Applicant Management
Applicant Filtering	Applicant Status Management
Interview	Selection Process
Student Admission	Student Account Creation
Temporary Credentials	Email System
Student Groups	Group Management

CP Mentorship	CP Mentor
Development Mentorship	Development Mentor
Attendance	Attendance Management
CP Progress	Progress Tracking
Development Progress	Progress Tracking
Weekly Tasks	Assignments
Assignment Submission	Student Assignment Feature
Assignment Evaluation	Mentor Grading
Communication	Announcements
Administrative Monitoring	Admin Dashboard
Student Monitoring	Student Dashboard
Mentor Monitoring	Mentor Dashboard
Final Project	Full-Stack Application

49. SYSTEM FEATURE SUMMARY

The major features of the Bootcamp Management System are:

Public Features

- Landing page
- Bootcamp information
- Applicant registration
- Login

Applicant Features

- Application submission
- Application status processing
- Interview selection

Student Features

- Login
- Dashboard
- Profile
- Progress
- Attendance
- Announcements
- Assignments
- Assignment submission
- Assignment status
- Grades

Mentor Features

- Login
- Dashboard
- Assigned students
- Student progress
- Attendance recording
- Assignment management
- Submission review
- Assignment grading

Admin Features

- Login
- Dashboard
- Applicant management
- Student management
- Mentor management
- Group management
- Mentor assignment
- Attendance overview
- Progress overview
- Announcement management
- Assignment management

50. ROLE-BASED ACCESS SUMMARY

Feature

Student

Mentor

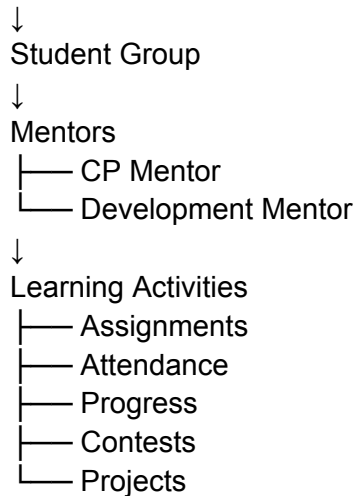
Admin

Login	Yes	Yes	Yes
View Own Profile	Yes	Yes	Yes
View Own Progress	Yes	Yes	Yes
View Assigned Student Progress	No	Yes	Yes
View Own Attendance	Yes	Yes	Yes
Record Attendance	No	Yes	Yes
View Assignments	Yes	Yes	Yes
Submit Assignments	Yes	No	No
Grade Assignments	No	Yes	Yes
View Announcements	Yes	Yes	Yes
Create Announcements	No	According to permission	Yes
Create Assignments	No	According to permission	Yes
Manage Applicants	No	No	Yes
Manage Students	No	No	Yes
Manage Mentors	No	No	Yes
Manage Groups	No	No	Yes
Assign Mentors	No	No	Yes
Attendance Overview	Own	Assigned Students	Overall
Progress Overview	Own	Assigned Students	Overall

51. SYSTEM DATA RELATIONSHIPS

The main relationships within the system can be represented as follows:

Applicant
↓
Student



These relationships allow the system to represent the actual structure of the bootcamp.

52. FINAL PROJECT SIGNIFICANCE

The ASTU MSJ Bootcamp Management System is significant because it is not simply a demonstration project created without a real use case.

It is designed around the actual needs and workflow of the ASTU MSJ Summer Bootcamp.

The system addresses several real challenges associated with managing a bootcamp, including:

- Managing applicants
- Filtering applicants
- Managing admission
- Creating student accounts
- Managing mentors
- Organizing student groups
- Tracking attendance
- Tracking progress
- Managing assignments
- Grading student work
- Communicating announcements
- Monitoring the overall bootcamp

The project therefore provides students with practical experience while also producing a system that can support the bootcamp itself.

